B

Bullock carts and bytes jostle together in the Sanand industrial belt on the outskirts of Ahmedabad. But don't be misled by the rustic settings. Ever since Tata shifted its massive Nano factory almost overnight from Singur in West Bengal in 2008, the township has flourished as a manufacturing hub. Now with three of the world's most sophisticated semiconductor chip factories being set up, Sanand is poised to become India's newest Silicon lodestone.

The shining example is the Murugappa Group's semiconductor chip venture CG Semi whose compact factory complex seems to have sprouted up from the ground as rapidly as the verdant rice fields that surround it. Just eight months ago, CG Semi set up its Rs 7,600 crore Outsourced Semiconductor Assembly & Test (OSAT) facility here in collaboration with Japan's Renesas Electronics Corporation and Thailand's Stars Microelectronics. On August 28, Ashwini Vaishnaw, the Union minister of electronics and information technology, flew down to inaugurate the brand new assembly pilot line that rolled out CG Semi's first made-in-India semiconductor chips.

A huge step forward, these chips are now being tested by clients to see if they match global industry standards. S. Vellayan, chairman, CG Power and Industrial Solutions Limited, of which CG Semi is a subsidiary, likened the challenge to being "just as steep as climbing Mount Everest". As he puts it, "Every chip we make here is a step towards India's technological sovereignty. It shows how government and industry can come together with conviction, capital and scale, and deliver rapidly."



"We took a leap of faith because chip making is a strategic and foundational industry, with a whole range of sectors depending on reliable supply apart from quality products"

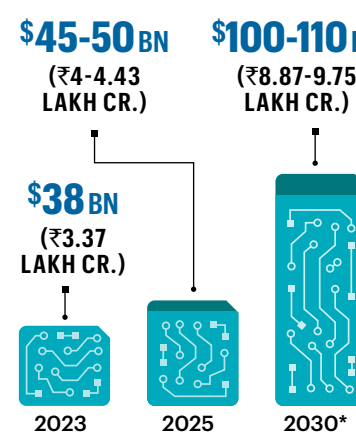
AMITESH SINHA | *CEO, India Semiconductor Mission*

Vellayan is not exaggerating. India needs 19 billion semiconductor chips annually. They are the heartbeat of a range of devices, be it smartphones, washing machines, refrigerators, cars or missiles and spaceships. But because the country lacks major semiconductor facilities, we are forced to import 95 per cent of our requirements from a handful of semiconductor chip companies that dominate the world. The price tag? A hefty Rs 1.71 lakh crore a year. That is two and a half

INDIA'S CHIP CRUNCH

The size of the Indian semiconductor market is projected to double in the next five years

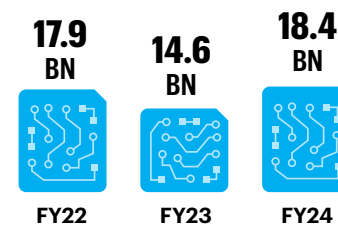
MARKET SIZE



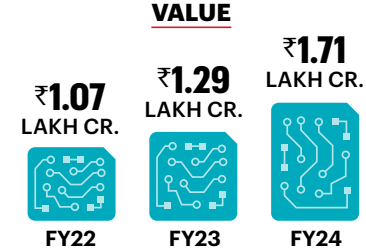
HEFTY IMPORT BILL

India imports about 95% of its chip needs, mainly from China, Taiwan, South Korea, Malaysia, Singapore and other Asian hubs

CHIPS IMPORTED



VALUE



Source: MeitY